

EAH Job Card and Tyre Request Process Documentation

Service App Controls, Workflow Logic, Purchase Order Linkages, and Management Oversight

Prepared for Management Review
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1. Executive Summary

The Service App controls maintenance spending through two main operational documents: **EAH Job Card** for vehicle maintenance and spare parts, and **Tyre Request** for tyre purchases and tyre maintenance. Both workflows connect to Purchase Orders while preserving traceability back to the originating operational document.

The system is designed to give management visibility over maintenance cost, prevent duplicate purchases, control item rates against approved Item Prices, and provide traceability for returned parts, worn-out tyres, and disposal.

2. Key Documents and Their Roles

Document	Purpose	Management Value
EAH Job Card	Records vehicle maintenance work, supplied parts, labour rates, supplier, project, cost center, and vehicle details.	Controls maintenance cost and gives a central view of parts, labour, warranty checks, and purchase linkage.
Purchase Order	Procurement document generated from approved maintenance or tyre requests.	Ensures payment is based on controlled operational documents and approved rates.
Maintenance Return Note	Records returned used parts from a submitted EAH Job Card.	Creates accountability for removed parts and prevents untracked reusable/scrap stock.
Used Spare Parts Issue Note	Issues previously returned spare parts for internal reuse or other purposes.	Tracks outflow of returned parts and prevents issuing more than available quantity.
Tyre Request	Captures tyre purchase or tyre maintenance requests for a vehicle.	Controls tyre replacement and maintenance spending, including worn-out tyre traceability.
Tyre Receiving Note	Records receipt of worn-out tyres after a tyre request.	Ensures replaced tyres are physically accounted for.
Tyre Disposal Note	Records disposal of received worn-out tyres.	Controls final disposal and prevents disposal quantities from exceeding received balances.

3. EAH Job Card Workflow

1. User creates an EAH Job Card for a vehicle maintenance event.
2. User selects required header information: vehicle, service date, supplier, price list, project, cost center, driver, and faults.
3. User adds supplied parts and labour-rate operations.
4. The system validates required fields, price controls, make compatibility, warranty/recent-service warnings, and totals.
5. When submitted, the user can create a Purchase Order and a Maintenance Return Note.
6. The Purchase Order carries the link back to the EAH Job Card for audit and duplicate-payment prevention.

3.1 Required Header Controls

Required Field	Control Logic
Vehicle	Mandatory. The Job Card cannot be saved without vehicle selection.
Service Date	Mandatory. Used for service history and warranty checks.
Supplier	Mandatory. Used for procurement and purchase order routing.
Price List	Mandatory. Determines approved Item Price rates.
Project	Mandatory. Routed to Purchase Order project.
Driver Name	Mandatory. Supports operational accountability.

3.2 Supplied Parts Logic

For each supplied part, the system requires item and quantity. It auto-fetches item name and keeps each row's price list aligned with the parent Job Card price list.

If the part rate is blank, the system tries to fetch the approved Item Price. Current source logic uses the general Item + Price List rate first, and supplier-specific Item Price only as fallback.

3.3 Item Price and Rate Controls

Control	What It Does	Why It Matters
Approved Item Price Required	If a user enters a rate for a supplied part without an approved Item Price, the system raises a warning/blocking message.	Prevents uncontrolled purchases outside approved price lists.
Rate Cannot Exceed Item Price	Entered rate cannot be higher than the approved Item Price for the selected Price List.	Prevents overpricing and unauthorized supplier increases.
Supplier Less Strict	General Item Price is considered first; supplier-specific price is only fallback.	Allows use of company-approved rates even when supplier-specific pricing is not maintained.

3.4 Labour Rate Controls

The Job Card calculates service charges from labour-rate rows. Labour rows are also checked against the vehicle/job-card make so that the selected service operation template matches the Job Card's make context.

If operation templates are scoped to a specific make, the system blocks mismatched labour templates.

3.5 Totals Calculated

Total	Formula
Total Quantity	Sum of supplied parts quantities.
Spares Cost	Sum of Qty x Rate for supplied parts.
Service Charges	Sum of labour-rate amounts.
Total VAT Exclusive	Spares Cost + Service Charges.

3.6 Maintenance Control Warnings

Warning	Logic	Action
Recent Vehicle Service	Checks whether the same vehicle was recently serviced within the configured service interval.	User receives warning; submit can be blocked unless Override Controls is checked.
Part Warranty	Checks whether a part may still be under warranty based on previous service date and item warranty period.	User receives warning; submit can be blocked unless Override Controls is checked.

3.7 Price Insight and Fluctuation Banner

The EAH Job Card has a **Price Insight** button. It checks submitted Purchase Orders in the last 90 days and compares each supplied part's current rate against historical rates.

Status	Meaning	Visual Signal
Normal	Price increase is below review threshold.	No critical action required.
Review	Current rate increased by 20% or more compared with recent history.	Orange alert for approver review.
High Risk	Current rate increased by 50% or more compared with recent history.	Red alert: Price Fluctuation Detected.
No History	No submitted Purchase Order history found in the last 90 days.	Blue alert for new/untracked item history.

The banner includes an **Open Price Insight** button that opens a detailed table showing current rate, last PO rate, change %, lowest/highest rate in 90 days, last supplier, last PO, and last purchase date.

Management note: the current source checkout contains the Price Insight and banner logic. The separate "Price Fluctuation Reason" fields were requested, but they are not visible in the current EAH Job Card JSON in this checkout. If those fields are deployed in live, they should be treated as the user justification area for price changes.

4. Purchase Order Creation from EAH Job Card

Purchase Orders are generated from submitted EAH Job Cards only. The generated Purchase Order carries the operational context so management can trace spending to the exact Job Card.

EAH Job Card Field	Purchase Order Field
Supplier	Supplier
Project	Project
Cost Center	Cost Center
Vehicle	Custom Vehicle field where available
EAH Job Card Name	EAH Job Card / Job Card Link compatibility fields
Supplied Parts	Purchase Order Items
Labour Charges	Additional labour item row, if configured and amount is greater than zero

4.1 Duplicate Purchase Order Control

The system checks whether an active Purchase Order already exists for the same EAH Job Card. If yes, it blocks creation of another Purchase Order to prevent duplicate payments.

4.2 Purchase Order Deletion/Cancellation Link Cleanup

When Purchase Orders linked to an EAH Job Card are cancelled or deleted, the system clears the EAH Job Card back-link so Frappe's link protection does not make document cleanup difficult.

5. Maintenance Return and Used Spare Parts Controls

5.1 Maintenance Return Note

After a submitted EAH Job Card, users can create a Maintenance Return Note to capture removed/returned spare parts.

Control	Purpose
Only submitted EAH Job Cards can create return notes	Prevents returns from unapproved work.
Only one active return note per Job Card	Prevents duplicate returns.
Qty received cannot exceed qty expected	Prevents over-receipting.
Status calculation	Draft, Partially Received, Fully Received, or Cancelled.

5.2 Used Spare Parts Issue Note

Returned spare parts can be issued later through a Used Spare Parts Issue Note. The system tracks available balances from Maintenance Return Notes and prevents issuing more than available.

Control	Purpose
Submitted return note required	Ensures source stock is confirmed.
Manual source requires manual reason and remarks	Improves auditability for opening/reconciliation entries.
Qty out must be greater than zero	Prevents empty issue documents.
Qty out cannot exceed available quantity	Prevents negative returned-part balances.

6. Tyre Request Workflow

Tyre Request supports two modes: **New Tyre Purchase** and **Tyre Maintenance**. The form behavior changes based on request type.

Request Type	Active Table	Typical Output
New Tyre Purchase	Tyre Items	Purchase Order and Tyre Receiving Note
Tyre Maintenance	Tyre Maintenance rows	Purchase Order for maintenance service

6.1 Required Header Controls

Required Field	Purpose
Vehicle	Identifies the truck/vehicle requiring tyres.
Request Date	Controls chronology and receipt checks.
Supplier	Mapped to Purchase Order.
Price List	Used for approved tyre Item Price lookup.
Project	Mapped to Purchase Order and item rows.
Cost Center	Mapped to Purchase Order and item rows.
Odometer Reading	Operational tracking for tyre usage.

6.2 New Tyre Purchase Controls

Control	Logic
At least one tyre row	The request cannot be saved without tyre items.
Item must belong to Tyres group	Only items under Item Group "Tyres" are allowed for tyre purchase rows.
Qty must be greater than zero	Prevents invalid or zero-quantity requests.
Wheel position required	Ensures each tyre is traceable to its vehicle position.
Worn-out serial and brand required	Captures the tyre being replaced for accountability.
Approved Item Price required	Tyre rate must exist in the selected Price List.
Rate cannot exceed Item Price	Prevents overpricing of tyre purchases.

6.3 Outstanding Receipt Control

For new tyre purchases, the system checks whether the same vehicle has an earlier submitted Tyre Request with missing or incomplete Tyre Receiving Note. If yes, it blocks the new request.

This ensures worn-out tyres from previous replacements are received/accounted for before another tyre replacement cycle starts.

6.4 Tyre Maintenance Controls

Control	Logic
At least one maintenance row	Tyre maintenance requests require operation rows.
Tyre Maintenance Item required	Used as the Purchase Order item for maintenance service.
Tyre position required	Identifies where the maintenance applies.
Operation required	User must select the FPSE/operation.
Rate must be greater than zero	Prevents zero-value maintenance requests.

6.5 Totals

Request Type	Total Quantity	Total Purchase Amount
New Tyre Purchase	Sum of tyre item quantities	Sum of Qty x Rate

Tyre Maintenance

Count of maintenance rows

Sum of maintenance row rates

7. Purchase Order Creation from Tyre Request

Purchase Orders can only be created from submitted Tyre Requests. The system blocks duplicate active Purchase Orders for the same Tyre Request.

Tyre Request Field	Purchase Order Field
Supplier	Supplier
Request Date	Transaction Date / Schedule Date
Project	Project on PO and PO Items
Cost Center	Cost Center on PO and PO Items
Tyre Request Name	Custom Tyre Request Link
Tyre Items	Purchase Order Items with tyre descriptions

7.1 Tyre Request Purchase Order Integrity

If a Purchase Order was generated from a Tyre Request, the mapped item rows and parent supplier, project, and cost center must remain identical to the Tyre Request. If changes are needed, users should update the Tyre Request and recreate the Purchase Order.

8. Tyre Receiving and Disposal Controls

8.1 Tyre Receiving Note

Control	Purpose
Only submitted Tyre Requests can create receiving notes	Prevents receipt against unapproved requests.
Only one active receiving note per Tyre Request	Prevents duplicate receipts.
Qty received cannot be negative	Prevents invalid receiving.
Qty received cannot exceed expected qty	Prevents over-receipt.
Status auto-calculated	Draft, Partially Received, Fully Received, or Cancelled.

8.2 Tyre Disposal Note

Control	Purpose
Submitted Tyre Receiving Note required	Disposal can only happen after receipt is confirmed.
Qty out must be greater than zero	Prevents empty disposal documents.
Qty out cannot exceed available quantity	Prevents disposing more tyres than received and undisposed.
Default disposal metadata	Posting date, disposed by, and disposal method are set automatically where blank.

9. Management Controls Summary

Risk	Control in Place	Document
Overpricing of spare parts or tyres	Rate cannot exceed approved Item Price.	EAH Job Card, Tyre Request, Purchase Order
Supplier price fluctuation	Price Insight compares current rate with submitted PO history and flags Review/High Risk.	EAH Job Card
Duplicate payment for same maintenance	Only one active Purchase Order allowed per EAH Job Card.	Purchase Order / EAH Job Card
Duplicate tyre PO	Only one active Purchase Order allowed per Tyre Request.	Tyre Request
Unaccounted worn-out tyres	New tyre request blocked if previous tyre receipt is outstanding.	Tyre Request / Tyre Receiving Note
Receiving or issuing excess quantities	Qty received/out cannot exceed expected or available balances.	Maintenance Return, Used Spare Issue, Tyre Receiving, Tyre Disposal
Wrong cost allocation	Project and Cost Center route to Purchase Orders and item rows.	EAH Job Card, Tyre Request
Wrong operation template	Labour operation make must match Job Card make where scoped.	EAH Job Card

10. Recommended Management Review Practice

1. Approvers should always review the EAH Job Card banner before submission or Purchase Order creation.
2. Where **Price Fluctuation Detected** appears, approvers should open Price Insight and compare current rate against recent history.
3. Finance/procurement should maintain Item Prices for frequently purchased spare parts and tyres.
4. Users should not bypass Price List controls by manually changing Purchase Order rates after mapping.
5. Tyre Receiving Notes should be completed immediately after replacement so new tyre requests are not blocked.
6. Returned parts and worn-out tyres should be physically reconciled against return/receiving/disposal notes.

11. Operational Checklist

Before Submitting EAH Job Card

- Vehicle, supplier, project, price list, and driver are filled.
- Supplied parts have approved Item Prices.
- Rates do not exceed approved Item Price.
- Price Insight is reviewed where banner appears.
- Warranty or recent-service warnings are resolved or override is justified.

Before Submitting Tyre Request

- Vehicle, supplier, price list, project, cost center, and odometer are filled.
- Tyre items belong to Item Group Tyres.
- Wheel position, worn-out serial, and worn-out brand are captured.
- Approved Item Price exists and rate is within allowed limit.
- Previous tyre receiving for the vehicle is complete.

Before Paying Purchase Orders

- Confirm Purchase Order is linked to the correct EAH Job Card or Tyre Request.
- Confirm project, cost center, and vehicle routing are correct.
- Confirm no duplicate Purchase Order exists for the same source document.
- Confirm item rows match the source operational document.